

6 (long)

A pin jointed 2D truss structure is shown in Fig. 6. The members of the truss have cross-sectional area A and Young's Modulus E such that $AE = 10^7 \text{ N}^{-1} \text{ m}$.

A vertical point load of 10 kN is applied at joint D.

- (a) Calculate the support reactions at A and C. [4]
- (b) Using graphical methods or otherwise, determine the member forces. [6]
- (c) Using graphical methods or otherwise, determine the horizontal displacement of joint C. [20]

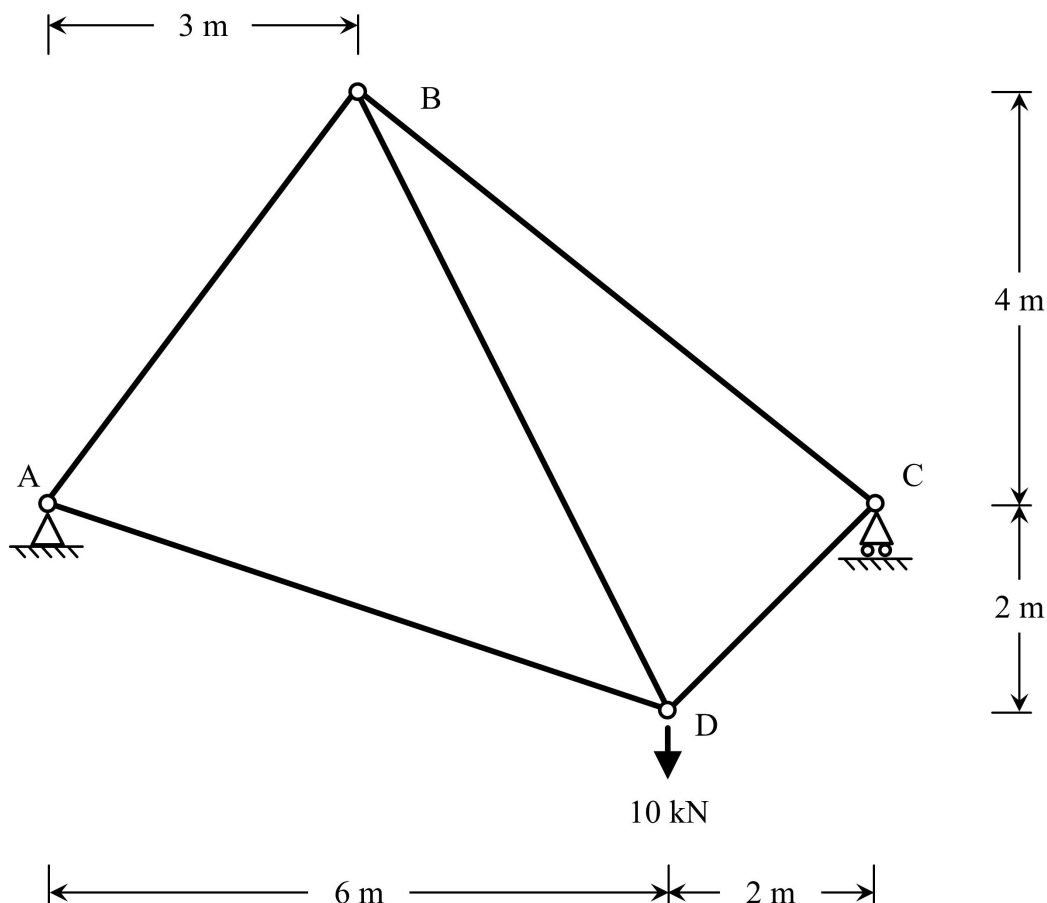


Fig. 6